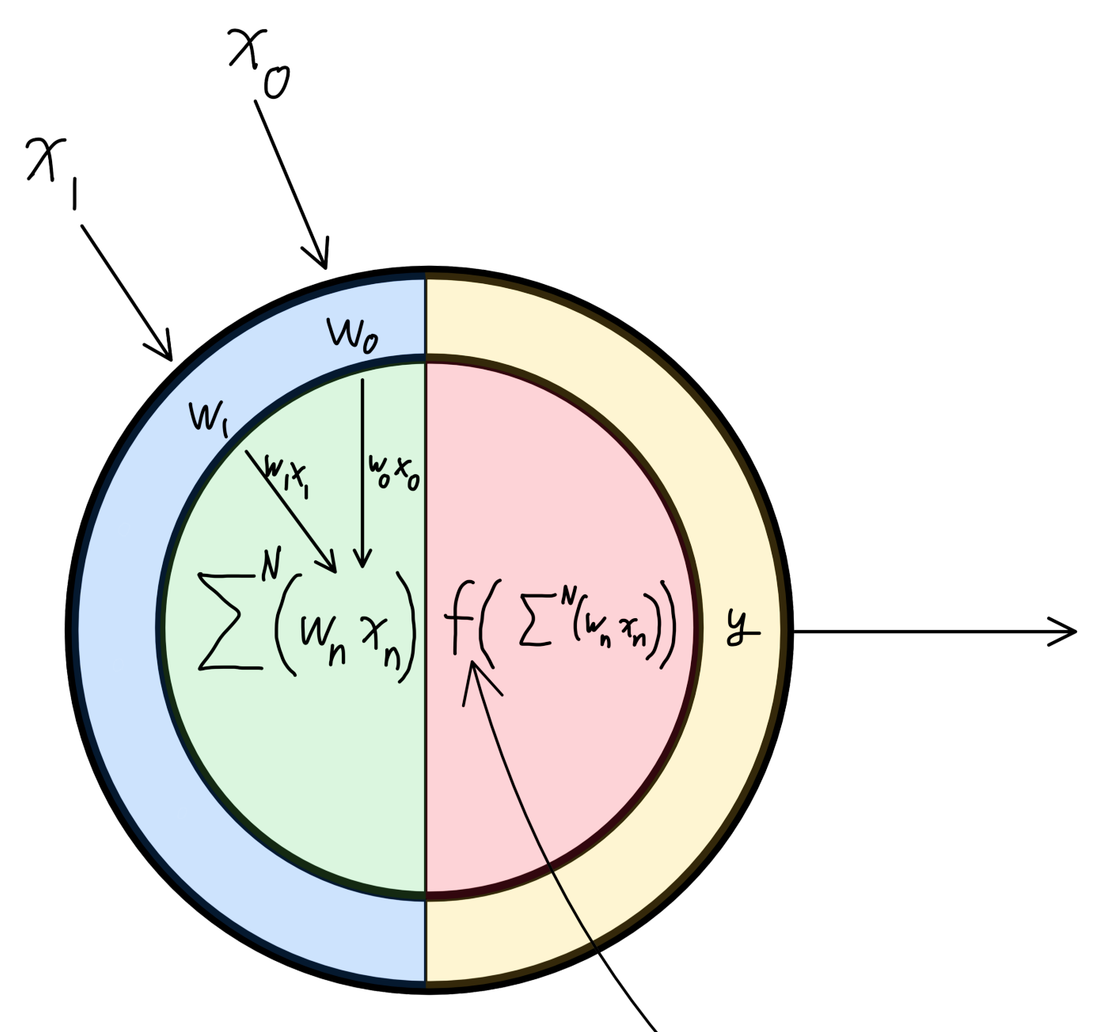
Linear Model



Gradient descent algorithm

$\nabla E(w(t)) = \frac{-1}{N} \sum_N \frac{y_n x_n}{1 + e^{(y w^T(t) x_n)}}$

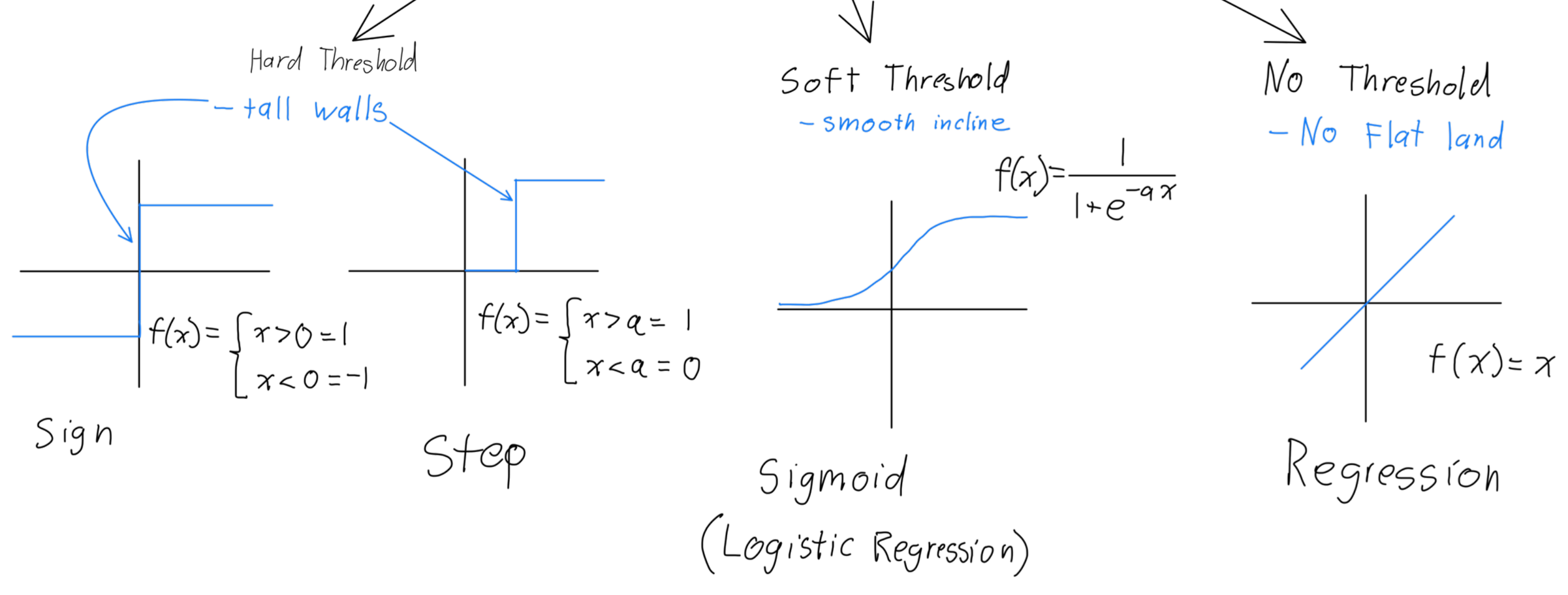
$\nabla E(w(t))$ = can't every reach 0 (asymptote at 0)

$w(t+1) = w(t) - \eta \nabla E(w(t))$

Activation Function f

input $\longrightarrow$ f $\longrightarrow$ output

activation



Hard Threshold

- tall walls

Soft Threshold

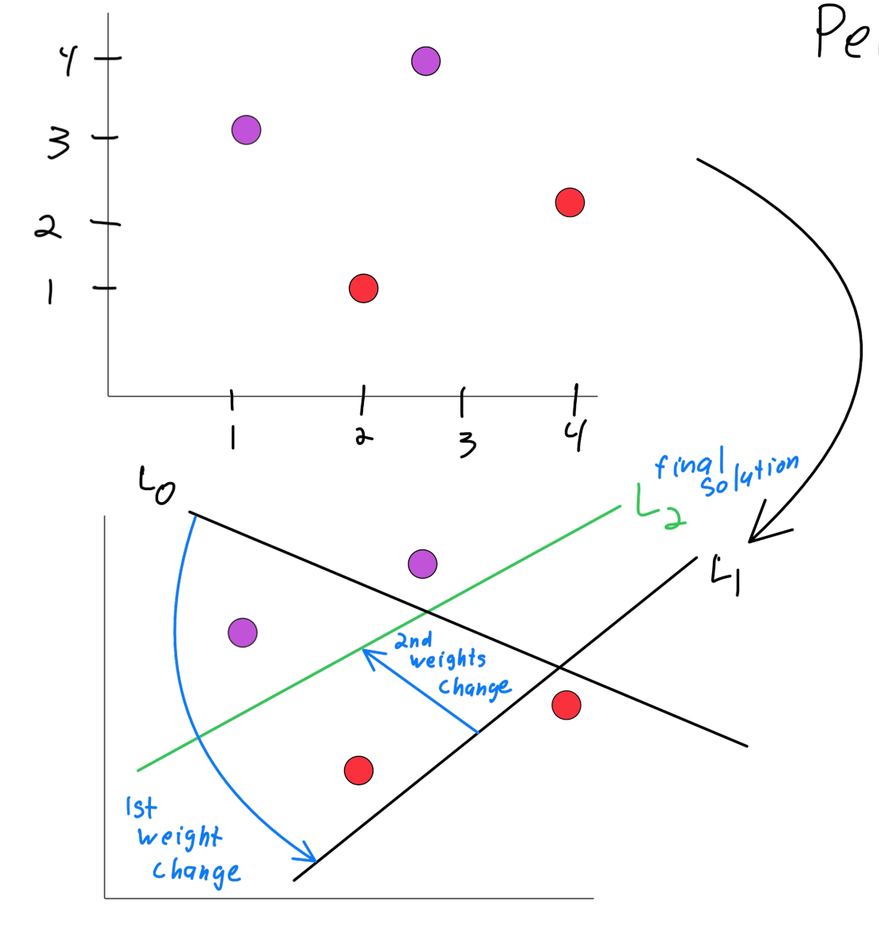
- smooth incline

No Threshold

- No Flat land

Linear classification

Linearly Separable Points



Perceptron Learning Algorithm (PLA

f = sign

Multilinear logistic regression with soft max

Logistic Regression approach can work with $K > 2$ classes

$P(y_n = k \mid x_n) = \frac{e^{w_k^T x_n}}{\sum_{i=1}^K e^{w_i^T x_n}}$ for $k = 1, 2, 3 \ldots K$

w_k is the vector M weights w_{mk} for class k

Bayes theorem

Joint probabilities $P(A, B) = P(B, A)$

Conditional Probabilities $\underbrace{P(B)}P(A|B) = P(A)P(B|A)$

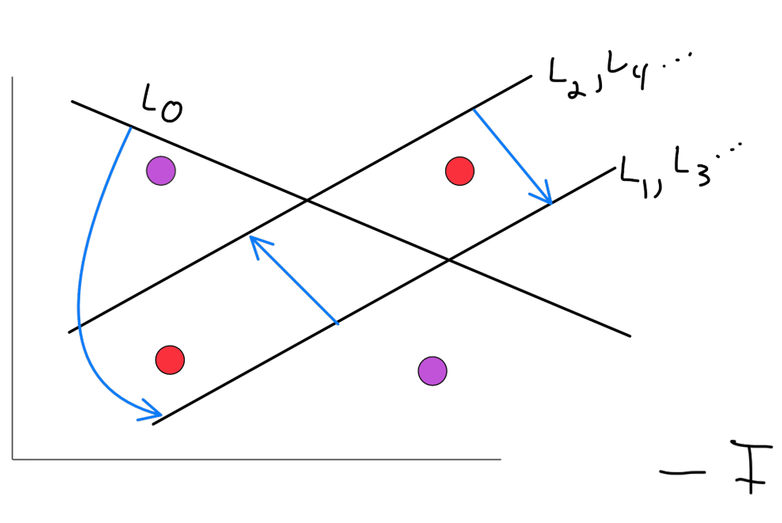
$\underset{\text{posterior}}{P(A|B)} = \frac{\overset{\text{prior}}{P(A)} \overset{\text{likelihood}}{P(B|A)}}{\underset{\text{evidence}}{P(B)}}$

Assuming Conditional independence

$p(x_n | y_n = k) = P(x_{n0} | y_n = k) \cdot P(x_{n1} | y_n = k) \cdot \ldots \cdot P(x_{nM} | y_n = k)$

$= \prod^M P(x_{nm} | y_n = k)$

But what about NonLinear Separable Points



PLA, f = sign

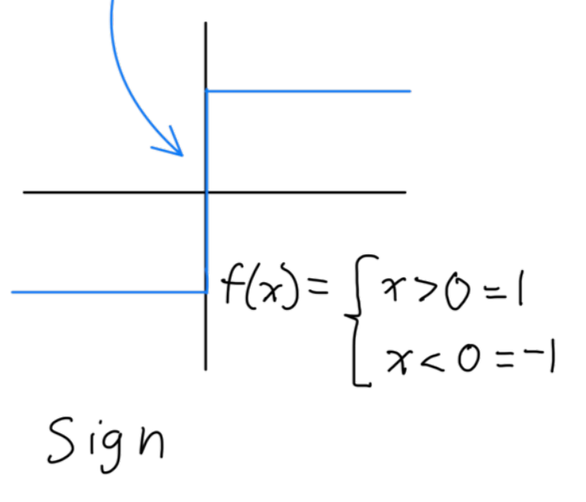
PLA ∞ alternates can't find solution

- Issue : using Hard threshold activation function
- Minimize sign's error function is NP-hard
- Need an alternative

Can't derive

hard threshold's functions aren't continous (the wall

what's sign(0) ???



Soft Threshold functions

$\theta(x) = \frac{e^x}{1+e^x}$

Logits

$\text{logit}(\theta) = x = \ln\left(\frac{\theta(x)}{1-\theta(x)}\right)$

Odds

$e^x = \frac{\theta(x)}{1-\theta(x)}$

why soft threshold?

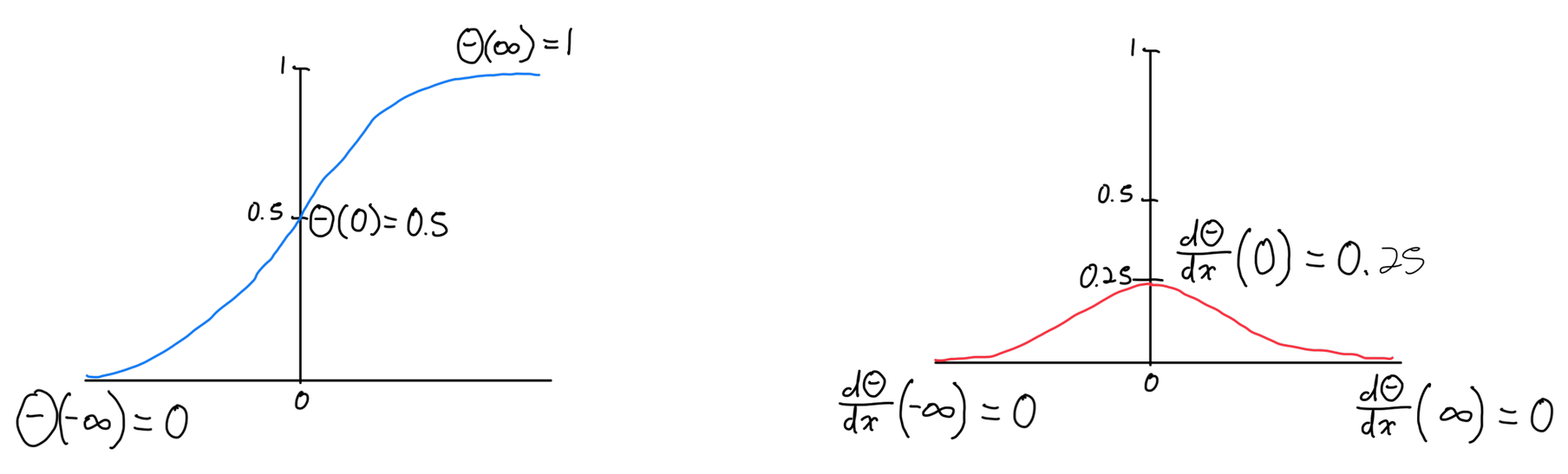
Logistic function

$\theta(x) = \frac{e^x}{1+e^x} = \frac{1}{1+e^{-x}}$

$\theta(-x) = 1 - \theta(x) = \frac{1}{1+e^x}$

$\frac{d\theta(x)}{d(x)} = \theta(x)(1-\theta(x))$

$= \theta(x)\theta(-x)$



* trade off

with $\theta(x)$ as error function there's no x where a minimum is located

Answers :

why we can't calc the minimum error like in PLA

Logistic Regression (Binary Classification)

$\theta(x)$ acts as a PDF (Probability Density Function)

Probability x_n is in category

$p(y_n = 1 | x_n) = \theta(w^T x_n)$

Probability x_n is not in category

$p(y_n = -1 | x_n) = 1 - \theta(w^T x_n)$

$= \theta(-w^T x_n)$

This binary why not 0 ???

$p(y_n = 1 | x_n) = \theta((1) w^T x_n)$

$= \theta(y_n w^T x_n)$

$p(y_n = -1 | x_n) = \theta((-1) w^T x_n)$

$= \theta(y_n w^T x_n)$

$p(y_n | x_n) = \theta(y_n w^T x_n)$

Maximum Likelihood and error function

What weights w maximizes likelihood from in-sample data

Multiply because the points are independent of each other ???

$L(w) = \prod_N p(y_n | x_n) = \prod_N \theta(y w^T x_n)$

why ???

$E(w) = -\frac{1}{N} \ln(L(w))$

$E(w) = -\frac{1}{N} \ln\left(\prod_N p(y_n | x_n)\right)$

$= \frac{1}{N} \sum_N \ln\left(\frac{1}{\theta(y w^T x_n)}\right)$

$= \frac{1}{N} \sum_N \ln\left(\frac{1}{\theta(y w^T x_n)}\right)$

$= \frac{1}{N} \sum_N \ln\left(\frac{1}{\frac{1}{1+e^{-y w^T x_n}}}\right)$

$E(w) = \frac{1}{N} \sum_N \ln(1 + e^{-y w^T x_n})$

Zero Error Gradient Impossible

$\frac{dE}{dw}(w) = -\frac{1}{N} \sum \frac{y x_n}{1 + e^{-y w^T x_n}}$

$= \frac{-1}{N} \sum \theta(y w^T x_n) y x_n$

There's no w where $\theta(y w^T x_n) = 0$ we can't find the exact minimum this way.

Gradient Error can be compute

$\frac{dE}{dw}(w) = \frac{-1}{N} \sum_N \theta(y_n w^T(t) x_n) y_n x_n$

$w(t+1) = w(t) - \eta \frac{dE}{dw}(w(t))$

- updating weights against the gradient's direction

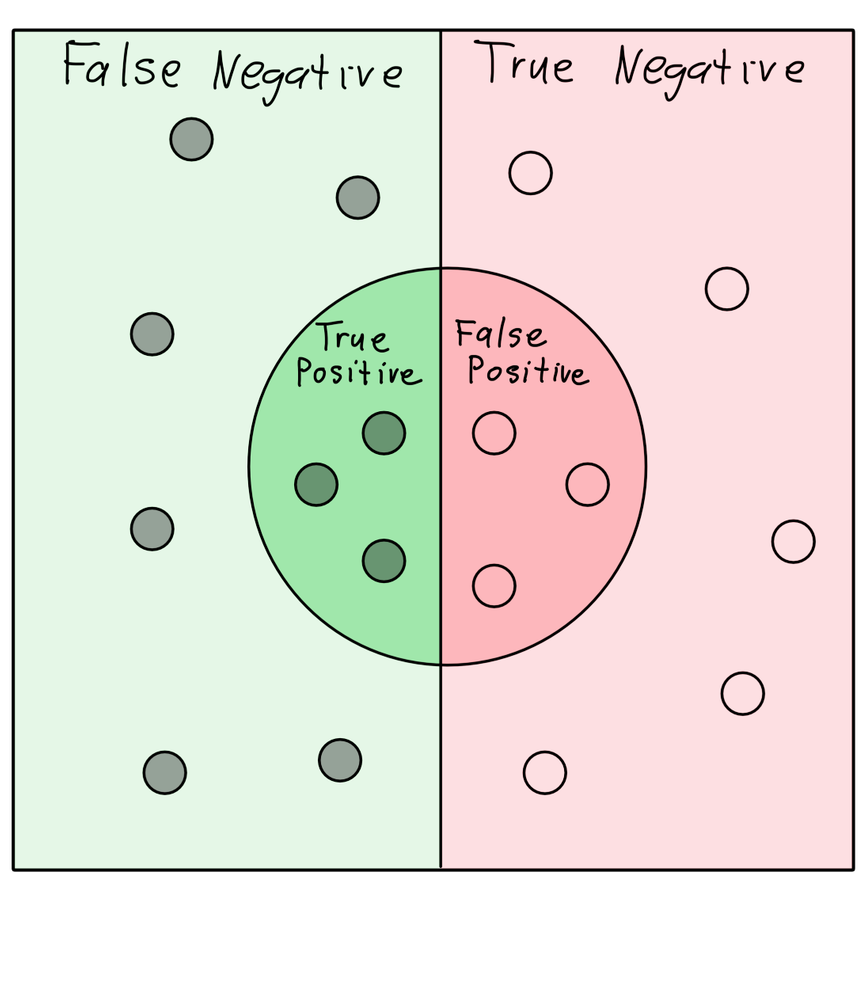
Soft Max

Creates Probabilities Distribution for the class

Model Evaluation

- Confusion Matrix

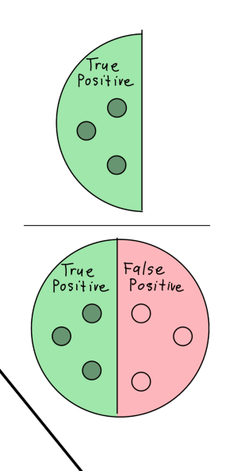
True Positive	False Positive
False Negative	True Negative



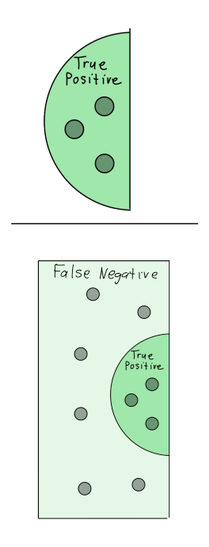
Support

occurance of each class

Percision =



Recall =



F_1 Score $= \frac{2}{\frac{1}{\text{Percision}} + \frac{1}{\text{Recall}}}$

- Somewhat Averages out Percision & Recall